House Price Prediction

Jeremy Guan Cheng-Cheng Liu Xudong Zhang Yixiao Li

1. Results

We evaluated multiple machine learning models for house price prediction using the Kaggle House Prices dataset, which contains 1460 training examples and 79 explanatory variables describing residential homes in Ames, Iowa. All models were trained using 5-fold cross-validation and evaluated using Root Mean Squared Error (RMSE) in log-space.

1.1. Model RMSE Comparison

Figure 1 summarizes the final RMSE performance across all implemented models.

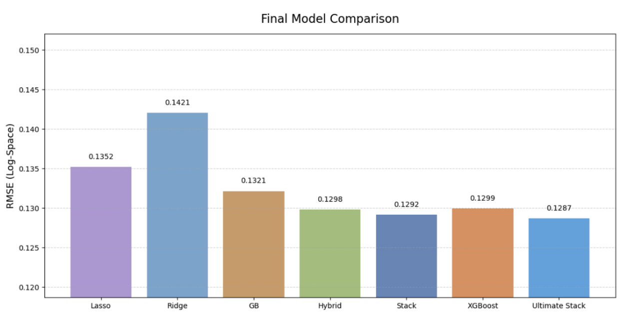


Figure 1. Final RMSE comparison across all models. Lower RMSE indicates better performance.

Among the linear models, Lasso Regression (Tibshirani, 1996) achieved slightly lower RMSE than Ridge Regression (Hoerl & Kennard, 1970) because L1 regularization performs implicit feature selection by shrinking less important coefficients to zero.

Nonlinear models performed substantially better overall. Gradient Boosting (Friedman, 2001) significantly outperformed the linear approaches, suggesting that housing prices contain strong nonlinear relationships that simple linear models cannot fully capture.

The Hybrid and Stacking ensemble methods (Wolpert, 1992; Breiman, 1996) provided additional improvements beyond standalone Gradient Boosting. XGBoost (Chen & Guestrin, 2016) achieved performance comparable to Gradient Boosting, while the Ultimate Stacking model achieved the best overall RMSE of 0.1287 by combining predictions from Lasso Regression, Gradient Boosting, and XGBoost.

Although the improvements from ensemble methods were relatively modest, the results suggest that combining more diverse nonlinear learners provides additional complementary predictive information.

1.2. Ultimate Stack Model

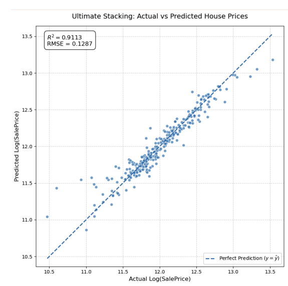


Figure 2. Ultimate Stacking predictions versus actual house prices in log-space. The dashed diagonal line represents perfect predictions.

Since the Ultimate Stacking model achieved the lowest overall RMSE among all evaluated approaches, we focus our detailed prediction analysis on this model. Presenting only the best-performing model provides a clearer visualization of overall prediction quality and remaining errors without introducing redundant plots for weaker models.

Figure 2 shows the prediction quality of the Ultimate Stacking model on the final test set. Most predictions lie close to the diagonal reference line, indicating a strong relationship between predicted and actual housing prices. The model achieved an R^2 score of 0.9113, indicating that the ensemble model successfully explained most of the variance in house prices.

Although most homes were predicted accurately, several outliers remain visible, particularly among extremely expensive homes. These prediction errors likely occur because luxury properties and rare neighborhood characteristics are underrepresented in the training data. In addition, some housing attributes may interact non-linearly in ways that are difficult for tabular features to fully capture. Overall, the prediction results demonstrate that ensemble methods generalized effectively and produced highly accurate predictions across a wide range of housing prices.

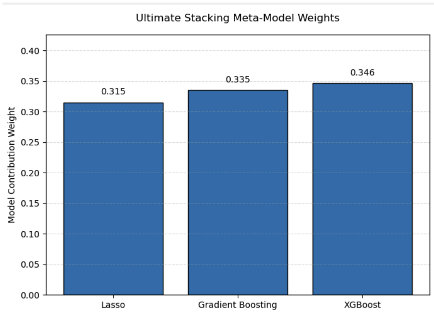


Figure 3. Meta-model weights learned by the Ultimate Stacking ensemble. Larger weights indicate stronger contribution to the final prediction.

1.3. Ultimate Stack Weights

Figure 3 shows the contribution weights learned by the stacking meta-model. XGBoost and Gradient Boosting received the largest weights, indicating that nonlinear tree-based learners contributed most strongly to the final predictions. Lasso Regression still provided complementary predictive information, suggesting that combining both linear and nonlinear modeling approaches improved overall ensemble performance.

1.4. Residual Analysis

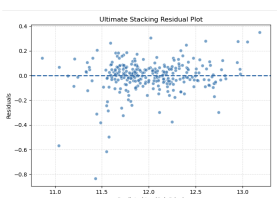


Figure 4. Residual plot of the Ultimate Stacking model. Residuals remain centered near zero, indicating limited systematic prediction bias.

Figure 4 shows the residual distribution of the Ultimate Stacking model. Most residuals remain concentrated around zero, indicating that the model does not exhibit strong systematic prediction bias. Larger residual variance appears primarily among expensive homes, suggesting that luxury properties remain more difficult to predict accurately due to rare or highly nonlinear housing characteristics. Manual inspection of several high-error predictions revealed that many poorly predicted homes contained uncommon feature combinations, including unusually large living areas, premium neighborhoods, and extensive renovations. These rare housing configurations were underrepresented in the training data, making accurate generalization more challenging for the ensemble model.

1.5. Discussion

Overall, the experimental results demonstrate that ensemble learning provides the strongest predictive performance for tabular housing datasets. While regularized linear models improved substantially over the baseline regression model, nonlinear boosting approaches captured more complex relationships between housing attributes and sale prices. Combining both linear and nonlinear learners within the Ultimate Stacking framework produced the most accurate and robust predictions across the full range of housing prices.

The results also highlight the importance of preprocessing and feature engineering. Missing-value handling, log transformation, one-hot encoding, and feature scaling all contributed to improving model stability and reducing overfitting. These findings suggest that both data preprocessing and ensemble learning are critical for achieving strong performance on structured real-world machine learning problems.

2. Process

In this section, we will discuss our works, explorations, and iterations that led to the final solution.

2.1. Data Pre-processing: From Naive Imputation to Stratified Strategy

Initially, we implemented a naive imputation strategy by replacing all missing values with the feature mean. However, iterative diagnostics revealed that this approach violated the underlying statistical logic, as extreme outliers frequently pulled the mean away from the true central tendency.

To resolve this, we transitioned to a skewness-based imputation mechanism: features with an absolute skewness greater than 1.0 were imputed with the median, while roughly symmetric features retained mean imputation. For categorical data, we moved beyond uniform mode imputation by incorporating domain-specific insights. We observed that high missing rates often signaled the physical absence of a facility (e.g., ”No Basement”) rather than random omission . Consequently, we pivoted to imputing ”None” for facility-related features and reserved mode imputation only for specific features like `Electrical`, where missingness was diagnosed as a random human error due to its extremely low frequency and high class dominance.

2.2. Model Evolution: Overcoming Baseline Pathologies

Our modeling process was highly iterative, driven by diagnosing and resolving successive algorithmic pathologies. We established our initial baseline using unregularized Ordinary Least Squares (OLS) regression. Instead of standard errors, we observed a catastrophic failure in predictive stability, yielding a sub-optimal R^2 score of 0.6797 and violent RMSE fluctuations across cross-validation folds.

By visualizing the coefficient weights, we discovered a severe case of algorithmic hijacking caused by highly collinear categorical variables. The OLS model created "predictive landmines" by assigning extreme negative weights (e.g., -3.6571) to sparse configurations. Lacking a penalty term, it suffered from perfect multicollinearity, arbitrarily distributing identical weights across related `_None` features.

Diagnosing this pathological behavior provided the definitive mathematical justification for our transition to regularized models (Lasso and Ridge). While L1 and L2 penalties successfully collapsed the inflated multicollinear weights, they introduced a new vulnerability: extreme sensitivity to feature scales and skewness, which we ultimately resolved using the Yeo-Johnson transformation.

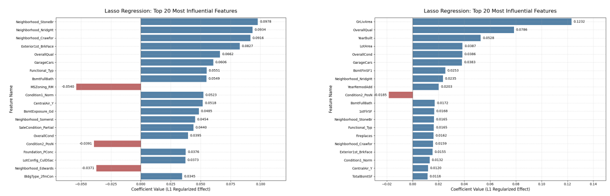
However, even after stabilizing the linear baselines, we recognized a fundamental limitation: linear models cannot inherently capture complex, non-linear interactions between housing features (e.g., the compounding value of high overall quality in a premium neighborhood). This realization catalyzed our architectural pivot to the Gradient Boosting Regressor (GBR). By utilizing an ensemble of decision trees, GBR fundamentally bypassed the multicollinearity trap, required less rigid assumptions about data distribution, and successfully captured the non-linear nuances of the real estate market. Ultimately, this successful transition from linear frameworks to tree-based ensembles not only improved predictive stability but also laid a robust foundation for our subsequent deployment of advanced algorithms, including XGBoost and Stacking architectures.

2.3. Feature Transformation: Resolving Weight Distortion

Upon applying regularized linear models (Lasso and Ridge), we observed a stark discrepancy in feature importance compared to our Gradient Boosting Regressor (GBR). The unstandardized linear models disproportionately inflated the weights of one-hot encoded categorical features, such as `Neighborhood`. In contrast, GBR logically prioritized fundamental structural metrics like `OverallQual` and `GrLivArea`.

We diagnosed this as a scale-induced distortion: because one-hot features are bounded between 0 and 1, the linear models assigned them massive coefficients to capture their impact on the target variable. To unify the scales, we initially applied standard scaling (`StandardScaler`). Surprisingly, this degraded the models' predictive performance. Further investigation revealed that standard scaling failed to correct the underlying high skewness of our numerical features, which violated the homoscedasticity assumption of linear models.

To resolve both the scale disparity and the non-normality, we implemented a significant engineering pivot by introducing the **Yeo-Johnson power transformation**. This robust technique successfully normalized highly skewed features while standardizing their scales. Consequently, the predictive accuracy of our regularized models surpassed the unstandardized baseline, and their feature importance rankings perfectly aligned with the core structural drivers identified by the GBR model.



(a) Before: Unstandardized (b) After: Yeo-Johnson

Figure 5. Comparison of Lasso feature importance before and after resolving scale distortion and non-normality. The transformed Lasso perfectly aligns with GBR's feature priority.

Table 1. Impact of Feature Transformation on Model Performance (CV RMSE)

Strategy	Baseline	Lasso	Ridge
Unstandardized	0.1657	0.1445	0.1426
StandardScaler Only	0.1636	0.1480	0.1437
Yeo-Johnson Transform	**0.1501**	**0.1323**	**0.1346**

2.4. Model Interpretability: Diagnosing Counterintuitive Coefficients

Even after the Yeo-Johnson transformation optimized our regularized models, a granular inspection of the feature importance revealed counterintuitive predictive behaviors. For instance, the linear models assigned a negative weight to feature `Condition2_PosN` (proximity to a park or greenbelt), paradoxically implying that beneficial neighborhood amenities decrease home value.

As empirical evidence of this vulnerability, we traced this specific coefficient back to its raw training instances (Table 2). We discovered that this algorithmic anomaly was driven by a combination of categorical sparsity and isolated outliers. Datapoint Id 524 represents a massive anomaly: despite boasting the maximum `OverallQual` score of 10 and a sprawling `GrLivArea` of 4,676 sq.ft., its sale price is severely depressed at $184,750.

Table 2. Diagnosis of Outlier-Induced Weight Distortion

Id	SalePrice	Neighborhood	OverallQual	GrLivArea	Cond.1	Cond.2
524	$184,750	Edwards	10	4,676	PosN	PosN
826	$385,000	NridgHt	10	2,084	PosN	PosN

Because the PosN condition is extremely sparse within the dataset, the global linear model was forced to disproportionately punish this specific feature weight to mathematically accommodate the localized price collapse of this single outlier.

This empirical finding exposed a residual vulnerability in global linear models: their extreme susceptibility to localized noise within sparse features. This diagnosis served as the final catalyst for our transition to advanced tree-based ensembles. Unlike linear models, algorithms like the Gradient Boosting Regressor (GBR) and XGBoost can effectively isolate such anomalous datapoints into deep, specific splits, preventing localized noise from distorting the global feature logic.

2.5. Hyperparameter Optimization: From Grid to Stochastic Search

The performance of a Gradient Boosting Regressor (GBR) is highly sensitive to its hyperparameter configuration. To systematically identify an effective model structure, we initially employed a **Grid Search Cross-Validation (GridSearchCV)**. The search space was defined across four critical dimensions governing the ensemble's learning dynamics and tree architecture (Table 3).

Hyperparameter	Search Space	Best Parameter
`n_estimators`	[500, 1000]	1000
`learning_rate`	[0.01, 0.05, 0.1]	0.05
`max_depth`	[3, 4, 5]	3
`subsample`	[0.8, 0.9]	0.9

Table 3. GridSearchCV parameter space and optimal configuration.

While GridSearchCV achieved a strong CV RMSE of 0.1264, it suffers from two critical limitations. First, its discrete nature creates 'blind spots,' risking failure to find the global optimum. Second, exhaustive search scales poorly; the high computational cost (180 fits for 36 configurations) severely constrains search boundary expansion.

To comprehensively address the "blind spot" and the combinatorial explosion of the discrete grid, we subsequently transitioned to **RandomizedSearchCV**. The true advantage of this stochastic optimization lies in its ability to sample from continuous probability distributions rather than being confined to a finite list of predefined values (Table 4).

Hyperparameter	Search Range	Best Parameter
`n_estimators`	[300, 1500]	1452
`learning_rate`	[0.01, 0.10]	0.0249
`max_depth`	[3, 8]	3
`subsample`	[0.6, 1.0]	0.6795
`min_samples_split`	[2, 20]	10
`min_samples_leaf`	[1, 10]	7

Table 4. RandomizedSearchCV parameter distributions and the identified optimal configuration.

Beyond achieving higher predictive accuracy, RandomizedSearchCV demonstrated superior computational efficiency. By intelligently sampling from continuous distributions, it located a more optimal global minimum in just 100 iterations, compared to the 180 fits required by the exhaustive Grid Search.

Optimization Strategy	CV RMSE	Test RMSE	MPE
GBR (Grid Search)	0.1264	0.1353	14.49%
GBR (Random Search)	**0.1249**	**0.1317**	**14.08%**

Table 5. Performance comparison between Grid Search and Randomized Search optimization.

As demonstrated in Table 5, RandomizedSearchCV undeniably yields superior predictive performance.

2.6. Architectural Synthesis: From Hybrid to Stacking

While the Gradient Boosting Regressor (GBR) demonstrated strong individual performance, it suffers from a fundamental limitation inherent to tree-based models: the lack of **extrapolation capability**. Because decision trees partition the feature space into constant leaf values, they cannot inherently predict values beyond the maximum price encountered in the training set. To mitigate this, we initially implemented a **Hybrid Ensemble** by blending our regularized Lasso model with GBR. Using cross-validation, we performed a grid search across a range of blending ratios (from 1:0 to 0:1) to identify the optimal weight distribution between linear trends and non-linear nuances.

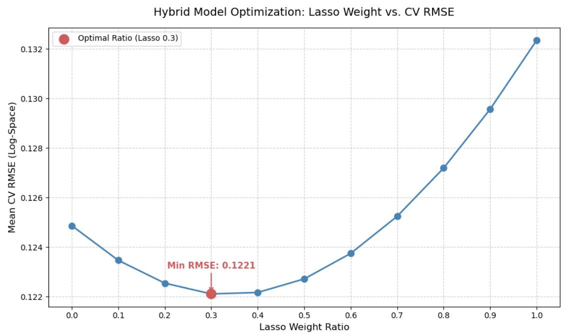


Figure 6. Cross-validation RMSE across different blending ratios for the Hybrid Ensemble.

Transitioning to a Stacking Architecture, we evaluated three configurations: (Lasso + Ridge + GBR), (Lasso + GBR), and (Ridge + GBR). The (Ridge + GBR) ensemble yielded the highest validation error (CV RMSE: 0.1250), whereas the lean (Lasso + GBR) combination matched the performance of the full three-model stack (CV RMSE: 0.1232). This demonstrates that Lasso's L1-driven feature selection provides a stronger, more complementary linear signal to the non-linear GBR than Ridge.

2.7. The Final Frontier: Integrating XGBoost

To further push the performance boundary, we integrated **XGBoost** into our ensemble. Unlike standard GBR, XGBoost incorporates advanced L1 and L2 regularization directly into its objective function, making it superior at handling the high-dimensional sparsity present in the Ames dataset. In our final architectural optimization, we pruned the redundant Ridge model and established a three-pillar Stacking framework comprising **Lasso, GBR, and XGBoost**. This synthesis successfully leveraged Lasso's global anchoring with the dual tree-based models' mastery of non-linear interactions, ultimately achieving our most robust and precise predictive performance (Test RMSE: 0.1287).

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